AN ELEMENT IS A UNIT EXACTLY WHEN ITS CLASS MODULO A LOCALLY NILPOTENT IDEAL IS A UNIT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Local nilpotence of an ideal. The hypothesis on I, used at proof lines 6510-6511. Gives z^N = 0 for each element z of the ideal, which is what terminates the geometric series argument.
- Congruence modulo a principal ideal. Used at proof lines 6509 and 6512. Lets the identity xy = 1 - z be read as 1 congruent to z modulo xR, so that x divides 1.

Lemma 0.1. *Let R be a ring and let $I \subset R$ be a locally nilpotent ideal. An element x of R is a unit if and only if the image of x in R/I is a unit.*

Proof. If x is a unit in R, then its image is clearly a unit in R/I. It remains to prove the converse. Assume the image of $y \in R$ in R/I is the inverse of the image of x. Then $xy = 1 - z$ for some $z \in I$. This means that $1 \equiv z$ modulo xR. Since z lies in the locally nilpotent ideal I, we have $z^N = 0$ for some sufficiently large N. It follows that $1 = 1^N \equiv z^N = 0$ modulo xR. In other words, x divides 1 and is hence a unit. □

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